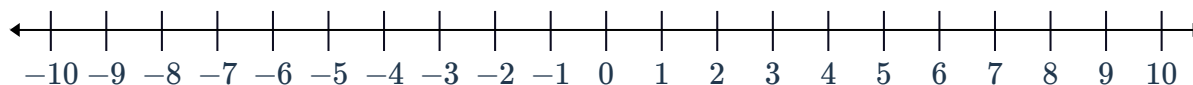


Absolute values



1 Consider the number line below:



- a How far is the number 3 from 0? b Hence, state the value of $|3|$.
c How far is the number -7 from 0? d Hence, state the value of $|-7|$.

2 Describe what the absolute value of -5 represents on a number line.

3 State whether the following have a value of $|-37|$:

A Both 37 and -37 **B** -37 only. **C** 37 only.

4 Determine whether the following have a value of 37:

A Both 37 and -37 **B** -37 only. **C** 37 only.

5 Evaluate the following:

- | | | | |
|-------------|------------|------------|-----------|
| a $ 6 $ | b $ 65 $ | c $ 3341 $ | d $ -3 $ |
| e $ -67 $ | f $ -155 $ | g $ 11 $ | h $ -21 $ |
| i $ -1000 $ | j $ -19 $ | k $ -756 $ | l $ 8 $ |

 6 Evaluate the following:

- | | |
|------------------------|-------------------------|
| a $ -2 + 5 $ | b $ -2 + 5 $ |
| c $ 8 - (-6) $ | d $ 8 - -6 $ |
| e $ -3 + -7 $ | f $ -3 + (-7) $ |
| g $ 8 - 9 + -6 + 4 $ | h $ 9 - 11 - -8 + 5 $ |

 **7** Evaluate the following:



a $|-3| \times |4|$

c $|10 \times (-2)|$

e $|-8| \times |-4|$

g $|6 - 8| \times |-7 + 4|$

i $\left| \frac{18 - 10}{3 - 7} \right|$

b $|-3 \times 4|$

d $|12| \div |-4|$

f $|-8 \times (-4)|$

h $\left| \frac{5 - 7}{-4 + (6)} \right|$

j $\frac{|6 - 12|}{3 - 4}$

8 If $|a| = 6$, what values could a be equal to? Explain your answer.

9 Find the possible values of x for each of the following:

a $|x| = 9$

b $|x| = 4$

c $|x| = 2$

d $|x| = 7$

e $|x| = 11$

f $|x| - 8 = 0$

g $|x| - 10 = 0$

h $|x| - 3 = 1$